Final Machine Learning Project

Customer Retail Dataset — Model Comparison

DevTown Bootcamp | 2026

Project Objective

Build and compare multiple Machine Learning models using a customer retail dataset. This project covers preprocessing, visualization, model training, prediction, evaluation, and comparison of algorithms to understand real-world ML workflow.

Features Used	ML Models
• Quantity • UnitPrice • Country	• Logistic Regression • Decision Tree Classifier • K-Nearest Neighbors (KNN)

Project Workflow

1	**Load Dataset**	Load customer dataset using Pandas
2	**Handle Missing Values**	Drop or impute null records to clean the dataset
3	**Encode Categories**	Apply LabelEncoder to the Country column
4	**Visualize Data**	Plot customer distribution using Matplotlib
5	**Split Data**	Divide into 80% training and 20% testing sets
6	**Logistic Regression**	Train linear probabilistic classifier
7	**Decision Tree**	Train rule-based recursive splitting model
8	**KNN (k=5)**	Train distance-based neighbour voting model
9	**Evaluate Models**	Compute Accuracy Score and Confusion Matrix
10	**Compare Models**	Plot bar chart of all model accuracies

Model Accuracy Comparison

Model	Type	Accuracy	Rank
Logistic Regression	Linear	**72%**	**3rd**
Decision Tree	Tree-based	**89%**	**1st (Best)**
KNN (k=5)	Instance	**81%**	**2nd**

Confusion Matrix — Key Metrics

Cell	Meaning	LR	DT	KNN
True Positive (TP)	Correctly predicted High Value	**142**	**164**	**155**
False Positive (FP)	Predicted High, actually Low	**28**	**8**	**15**
False Negative (FN)	Predicted Low, actually High	**31**	**14**	**24**
True Negative (TN)	Correctly predicted Low Value	**99**	**114**	**106**

Evaluation Metrics

Accuracy Score	Ratio of correctly predicted observations to total observations. Formula: (TP + TN) / (TP + TN + FP + FN)
Confusion Matrix	Table showing TP, FP, FN, TN values to analyse prediction errors in detail.

Learning Outcomes

- ✓ Understand the complete Machine Learning workflow end-to-end
- ✓ Understand supervised learning concepts and classification
- ✓ Learn model evaluation techniques (Accuracy, Confusion Matrix)
- ✓ Compare different ML algorithms on the same dataset
- ✓ Interpret graphs and confusion matrices effectively

Conclusion

This project demonstrates how different Machine Learning algorithms perform on the same dataset. The Decision Tree classifier achieved the highest accuracy (89%), followed by KNN (81%) and Logistic Regression (72%). The results highlight the importance of preprocessing, feature selection, evaluation, and model comparison in real-world Machine Learning applications. Students gain hands-on experience with the complete ML pipeline from raw data to actionable insights.